

Enabling High-Reliability Link Monitoring in In-Vehicle PONs via Correlation OTDR

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Abstract: We propose an in-vehicle PON monitoring scheme based on correlation OTDR. By incorporating in-band communication, the scheme achieves centimeter-level spatial resolution while enabling the detection of nanosecond-level outages and sub-0.2 dB power fluctuations. ©2026 The Author(s)

Introduction

As vehicles evolve toward higher intelligence, In-Vehicle Networks (IVNs) demand robust bandwidth and minimal latency for seamless sensor integration. Optical fiber surpasses copper in capacity, mass, and electromagnetic immunity, thereby driving extensive investigation of in-vehicle optical communications, including ongoing field trials and standardization [1,2]. Fiber-optic Ethernet and Passive Optical Networks (PONs) are the most prominent architectures; specifically, PONs leverage a P2MP topology to streamline zonal cabling. Nevertheless, IVNs must withstand rigorous automotive environments (characterized by temperature extremes of -35 to 85°C, vibration, and pollutants) that are significantly harsher than controlled data center environments. Consequently, ensuring automotive-grade reliability while maintaining high performance remains a critical research focus [1].

Vehicular sensing and control demand guaranteed delivery, necessitating real-time

monitoring and rapid fault intervention. Existing PON standards [2] are optimized for broadband access rather than vehicular systems, where microsecond-level outages are disruptive. Current schemes, hampered by kHz-frequency monitoring and minute-level statistics, fail to detect ns-level interruptions (Fig. 1). Thus, high-frequency physical-layer monitoring is essential for identifying impairments before critical power degradation.

While Optical Time Domain Reflectometry (OTDR) is the established technique for point-to-point link diagnosis [2], it is hindered by overlapping reflections in P2MP PONs and remains unsuitable for short in-vehicle fibers (<10 m) due to meter-level dead zones and insufficient spatial resolution. Alternative approaches, such as multi-wavelength scanning [4] and low-frequency PN sequence superposition [5], are constrained by high cost, temperature sensitivity, or inadequate resolution. Despite recent advances in machine learning for fault localization [6], to the best of our knowledge, no existing methodology has been specifically developed to address ns-level transient interruptions and real-time physical-layer power monitoring.

In this paper, we propose an in-vehicle PON monitoring scheme leveraging correlation OTDR (C-OTDR). By embedding pseudo-random noise (PN) sequences within in-band data frames as probing signals, the OLT correlates reflections with transmitted sequences to generate distinct correlation peaks. These peaks serve as high-frequency "heartbeats" that characterize real-time link status and power fluctuations. The system is capable of detecting ns-level transient outages, locating breakpoints with cm-level precision, and tracking power fluctuations below 0.2 dB. Such dynamic monitoring ensures the reliability of both the optical link and transceivers. Furthermore, by utilizing existing PON hardware and protocols, this method is highly cost-effective and facilitates future standardization.

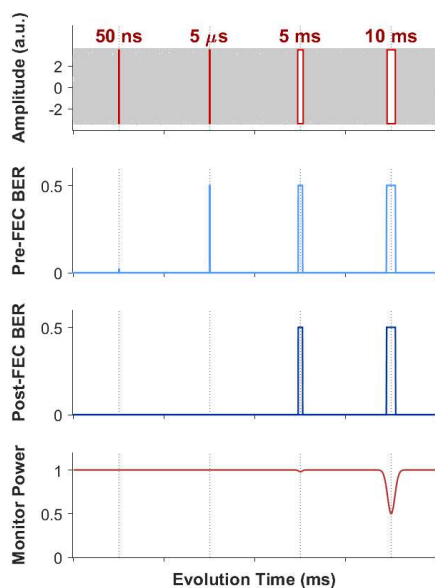


Fig. 1: Performance of optical transient interruption monitoring using various methodologies.

Principle

The proposed monitoring scheme is based on our prior C-OTDR system [7] using Direct Sequence Spread Spectrum (DSSS). The OLT transmits long PN sequences instead of short pulses, the receiver then correlates and de-spread the reflected signals with the original PN sequence. The correlation trace is derived by correlating the transmitted PN sequence $x_1(t)$ with received signal $x_2(t)$. The cross-correlation function is $R(\tau) = E[x_1(t) \cdot x_2(t - \tau)]$, where E denotes the expectation operator. A correlation peak $R_{max}(\tau)$ indicates alignment after delay τ . The reflection distance is calculated as $\Delta L = (c \cdot \tau)/2n$, where c/n represents the speed of light in the fiber. Since the correlation magnitude vanishes beyond a 1-bit offset, the spatial resolution is determined by the physical fiber length per bit. For instance, a 10-GS/s transmission rate theoretically yields a 1-cm spatial resolution.

In accordance with GPON standards, data payloads undergo multi-layer processing to form encapsulation frames. By reverse-engineering this encapsulation process, specifically tailored payloads can be constructed such that the resulting GPON frames exhibit a partial match with a target PN sequence. Due to the inherent auto-correlation properties of PN codes, these incomplete or partially modified sequences retain the ability to yield distinct correlation peaks upon cross-correlation processing. This approach enables precise synchronization within cluttered signals, ensuring high detection reliability despite signal interference, truncation, or erasure.

Experimental Setup

The experimental setup is illustrated in Fig. 2(a). A 1490-nm optical carrier from an External Cavity Laser (ECL) is modulated by a Mach-Zehnder Modulator (MZM), driven by an Arbitrary Waveform Generator (AWG) to emulate OLT downstream broadcasting. To regulate attenuation or induce transient optical interruptions, an Acousto-Optic Modulator (AOM) driven by an Arbitrary Function Generator (AFG) is cascaded in the feeder fiber. The downstream signal enters the Optical Distribution Network (ODN) as a probe, with 50% of the power received at the Optical Network Unit (ONU) and the remainder reflected back. After isolation via a circulator and an optical filter, the reflected and upstream signals are converted by Avalanche Photodiodes (APDs) and captured by a Digital Phosphor Oscilloscope (DPO) for offline analysis. The AWG generates periodic PN sequences with a polynomial of $G(x) = x^{14} + x^{13} + x^8 + x^4 + 1$ (length $2^{14}-1$) at 10 GS/s, while the DPO performs sampling at 25 GSa/s. Optical power levels at key monitoring points are summarized in Table 1.

In contrast to conventional OTDR, which fails to discriminate between overlapping branch reflections (Fig. 2(b)–(c)), the proposed scheme utilizes cross-correlation to resolve individual ONU paths. The resulting correlation peaks shown in Fig. 2(d) provide a precise mapping of fiber delays with a 2 cm resolution. By leveraging the robust auto-correlation properties of PN codes to eliminate inter-branch interference, this method enables stable, position-based ONU identification that is immune to environmental fluctuations,

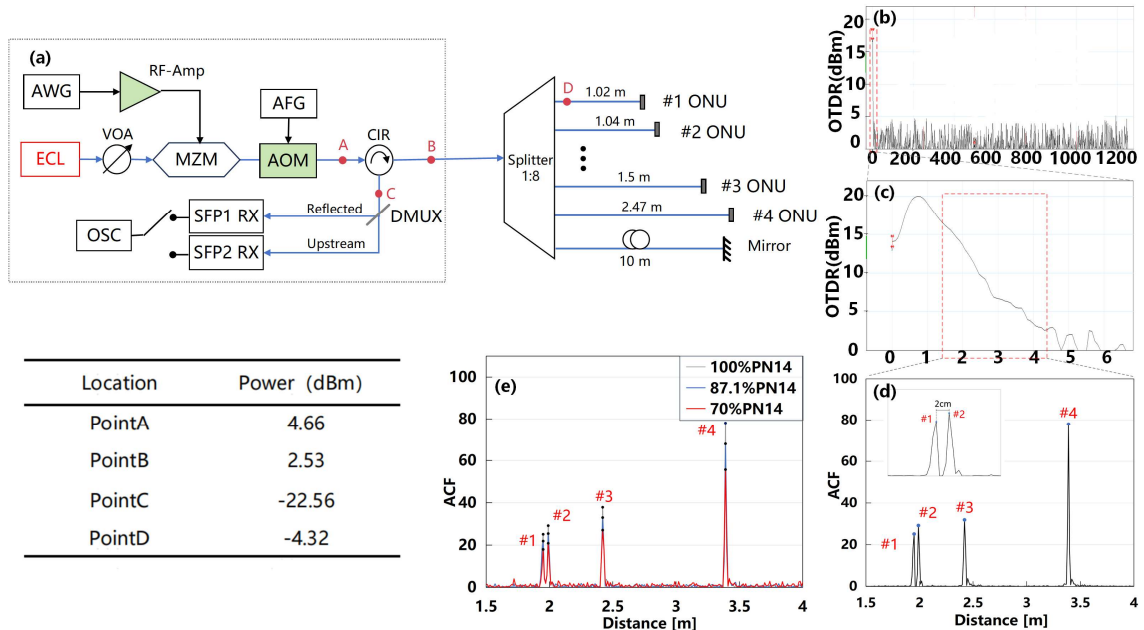


Fig. 2: (a) Experimental setup for C-OTDR-based link monitoring. (b) Overlapping ODN reflections captured by conventional OTDR and (c) a magnified view of the end-link trace. (d) Discrete ONU reflection peaks resolved via correlation OTDR. (e) Evolution of correlation peaks under varying PN-sequence alteration ratios.

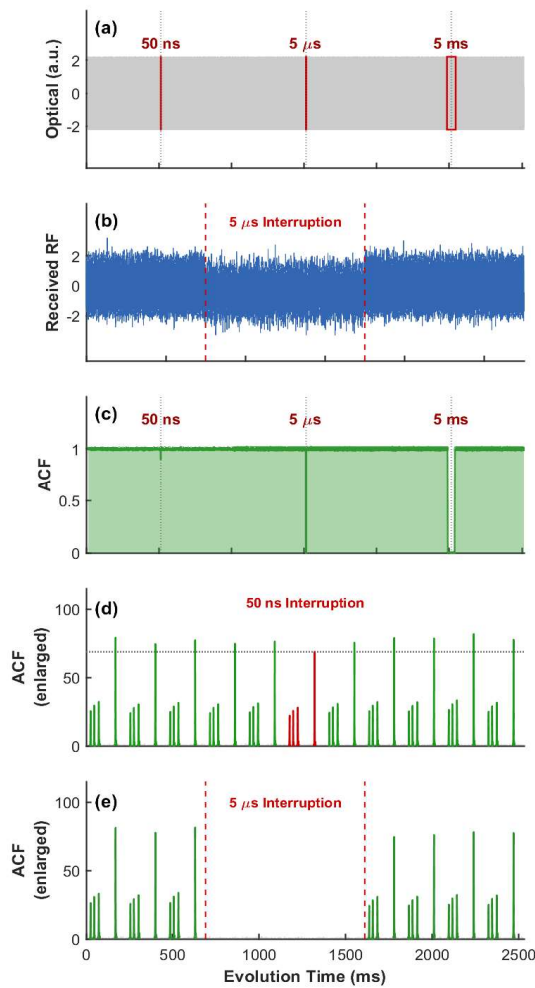


Fig. 3: (a) Transient optical outages of different durations, with their (b) received electrical signals and (c) resulting ACF envelopes. (d) and (e) detailed views of the ACF response to 50-ns and 5-μs interruptions.

such as temperature variations and mechanical stress.

The system's response to optical interruptions (50 ns, 5 μs, and 5 ms) was characterized using an AOM. While the electrical signal remains relatively stable during these outages (Fig. 3(b)), the ACF peak envelope shows distinct partial losses (Fig. 3(c)). Specifically, a 50 ns interruption induces a noticeable drop in the ACF peak due to fragmented PN sequences (Fig. 3(d)). For interruptions exceeding the 1.638 μs PN cycle, such as the 5 μs case in Fig. 3(e), approximately four PN sequences are erased, causing the total loss of the associated correlation peaks. These results highlight a superior detection sensitivity capable of capturing sub-microsecond transient events, a marked improvement over conventional techniques.

Fig. 4 demonstrates ACF peak analysis for detecting fiber outages and power fluctuations. In Fig. 4(a), distinct ACF peaks (red, green, blue,

and purple) allow for individual branch identification. Compared to the baseline, the disappearance of the blue peak and the decay of the red peak precisely indicate ONU1 disconnection and ONU2 attenuation, respectively. Furthermore, Fig. 4(b) shows the system's sensitivity to dynamic changes; the ACF peaks faithfully track a 0.2-dB, 100-kHz sinusoidal modulation induced by an AOM. These results confirm that the temporal evolution of ACF peaks provides a reliable metric for diagnosing transient outages and subtle power fading in vehicular PON environments.

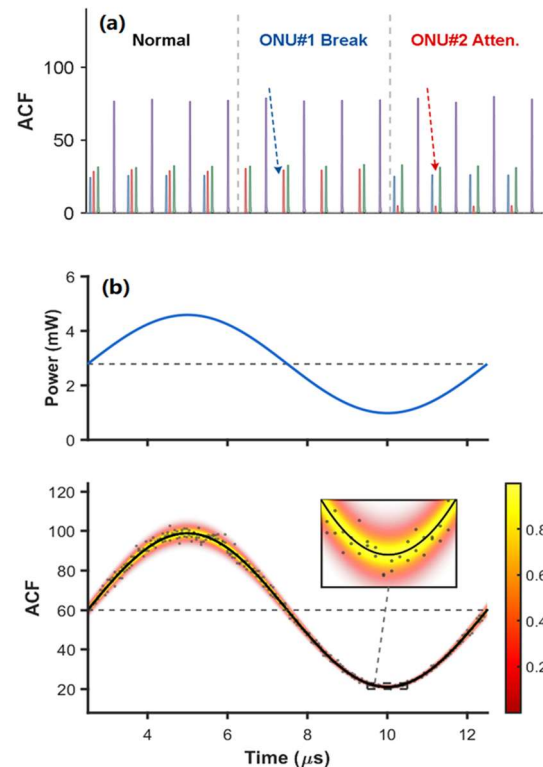


Fig. 4: (a) ACF peak variations under branch-specific attenuation and outages. (b) Sensitivity of ACF peak tracking to transient power fluctuations.

Conclusions

We have proposed and experimentally demonstrated an in-vehicle PON monitoring scheme based on correlation OTDR, utilizing in-band data for non-intrusive link sensing. The system effectively locates reflections with cm-level precision, detects ns-level interruptions, and tracks power dynamics under 0.2 dB. By enabling concurrent real-time tracking of link integrity and transceiver health, this approach provides a scalable and standardized framework for maintaining the operational stability of next-generation high-speed vehicular optical backbones.

Acknowledgements

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